

1. Research

1.1 Research plan

[The main requirement for the research is the precise definition of the research questions – goals of the research. These are the sources of the assumptions made before the experimentation.]

[This definition has to have a form that allows to verify whether the goal was achieved or not.]

[The reasoning for goals and other related premises have to be presented in this section.]

[The contents of the research plan have to be based on the main goal of the project.]

1.2 Datasets

[This section should describe the datasets used for the experimentation. The description should contain:

- Dataset origin – a paper, a webpage, an institution, etc. giving access to the dataset. The source should be presented in the form of a reference to the paper/hyperlink/institution's name,
- Dataset description – characteristics of the dataset such as:
 - number of samples,
 - number and types of features (real values, integer values, nominal values, strings etc.),
 - missing values,
 - A five-number summary of the data which is: the smallest data value (MIN), the first quartile (Q1), the median (MED), the third quartile (Q3), the largest data value (MAX): **[MIN, Q1, MED, Q3, MAX]**,
 - Identify outliers (if needed). Use the rule: a data point is an outlier if it is more than $1.5 \cdot \text{IQR}$ (Interquartile Range) above Q3 or below the Q1. Low outliers are below $Q1 - 1.5 \cdot \text{IQR}$ and high outliers are above $Q3 + 1.5 \cdot \text{IQR}$



- Dataset pre-processing – description of dataset transformations (if needed):
 - record removal,
 - normalization (value transformations e.g. scaling),
 - missing values filling policy (feature/record removal, filling with mean, median, mode, etc.),
 - dataset augmentation (new sample generation),
 - dataset division (e.g. for training, validation and testing),

- other.]

1.3 Research environment

[This section characterizes the execution environment of the implementation. Factors such as implementation language(s), libraries used and hardware configuration should be included in this designated section.]

1.4 Measures

[Measures used for describing results of experiments have to be defined in a separate section. Definition of a measure has to contain description of symbols used in it.]

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[The following sections represent groups of experiments. Those groups usually correspond to the research questions defined in the research plan.]

.....

1.5 [Experiment group section name]

Experiment 1. [Experiment title]

[All experiments have to have identical description style following this template. The experiment numeration does not reset for every group – experiments have unique numbers.]

Research question

[This part contains definition of the reason for the experiment.]

Assumptions

[Assumptions part has to show which hyperparameters were constant and which were changed during the experiment. Assumptions should be shown in form of a table or a list.]

Course of the experiment

[This part contains descriptions of: datasets or fragments of datasets used, step-by-step experimentation procedure and result evaluation measure. This and the previous parts of the description should allow a person to recreate the experiment.]

Results

[Results in form of: tables, graphs, screenshots, etc. are shown here.]

[A table caption has to be on the top of the table. A figure caption has to be under it.]

[Figures that represent graphs have to have: titles, descriptions and units of axes, and legends if needed.]

Comments

[Summary of the experiment in form of a commentary on presented results. Analysis and explanation of observed patterns, effects etc.]

Experiment 2. [Experiment title]

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1.6 [Experiment group section name]

Experiment 3. [Experiment title]

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1.7 Summary of research

[This final section summarizes the whole research. The results of all of experiments should be recalled there. The research summary should be done from the perspective of the main goal of the project.]

[Final note: if a figure/table is placed in the document, it has to be referred to in the text. Otherwise, it is considered an error. The same thing happens with the bibliography and citations. The only exception to this rule is in the “Results” part of experiments, where all the most important results have to be shown. If only the most important results are placed in the document, an additional file containing all the results must be provided.]